

CINDER v1.1.2 actuator dynamics — commercial-secondary recap

This note packages the current actuator-dynamics conclusion so the work can be rounded off for now and revisited later. The goal was not to prove a complete Sidewinder simulation. The goal was to show, in a defensible way, that the retained secondary dynamic terms need not be small and can produce a tangible whole-CVT trajectory difference.

The commercial hardware anchor remains a Yamaha Sidewinder style secondary with Dalton YSR helix options. The current source-backed component estimate gives reflected axial inertia levels from $0.45\times$ Baja (low) to $2.92\times$ Baja (high), with the nominal 31° case at $1.49\times$ Baja and the aggressive 28° angle sensitivity reaching $3.72\times$ Baja in the high estimate.

The final system-level study used a mechanism transplant: only the secondary helix geometry and movable-member rotational inertia were substituted into the otherwise frozen Baja host. A preregistered disturbance screen selected the strongest clean continuous nominal case, which was a $-120\text{ N}\cdot\text{m}$ added secondary torque ramped over 25 ms. That case did not reach the desired 10–20% nominal band, but it honestly landed at 7.87% while remaining mechanically clean and with the quasi-static helix torque magnitude never dropping below 92.2% of onset.

That selected nominal case still produced a visible trajectory consequence: max dynamic-vs-QS perturbation-response difference of 0.274 mm in shift, 40.4 rpm in primary speed, and 109.1 N in secondary clamp response. Under the same disturbance, the high estimate rose to 13.60% and 68.5 rpm.

Bottom line: the Baja machine remains close to quasi-static during ordinary continuous operation, but OTS-scale secondary hardware can move the retained helix dynamic correction out of the negligible regime. The full CVT prediction then changes measurably.

Interpretation boundary: this is a mechanism-transplant demonstration, not a reconstructed Sidewinder vehicle prediction.

Commercial scaling summary

Estimate	$I_{s,M}$ [kg m ²]	H@31° [rad/m]	$IH^2@31^\circ$ / Baja	$IH^2@28^\circ$ / Baja
low	0.00556	27.7	0.45×	0.57×
nominal	0.01288	33.3	1.49×	1.90×
high	0.02047	37.0	2.92×	3.72×

The key scaling story is that the commercial secondary can have weaker helix motion ratio H than Baja yet still exceed Baja in reflected axial inertia $I_{s,M} H^2$ because the movable-member rotational inertia is much larger.

Selected trajectory case and replay summary

Selected case	Added secondary torq	Ramp [ms]	Response class	Peak Pi_s, Mtra	tau_h, QS Peak	ΔF_helix	Selection status
s50_m120_r25	-120	25	clean_continuous	7.87%	92.2%	target_band_unreached_fallback	

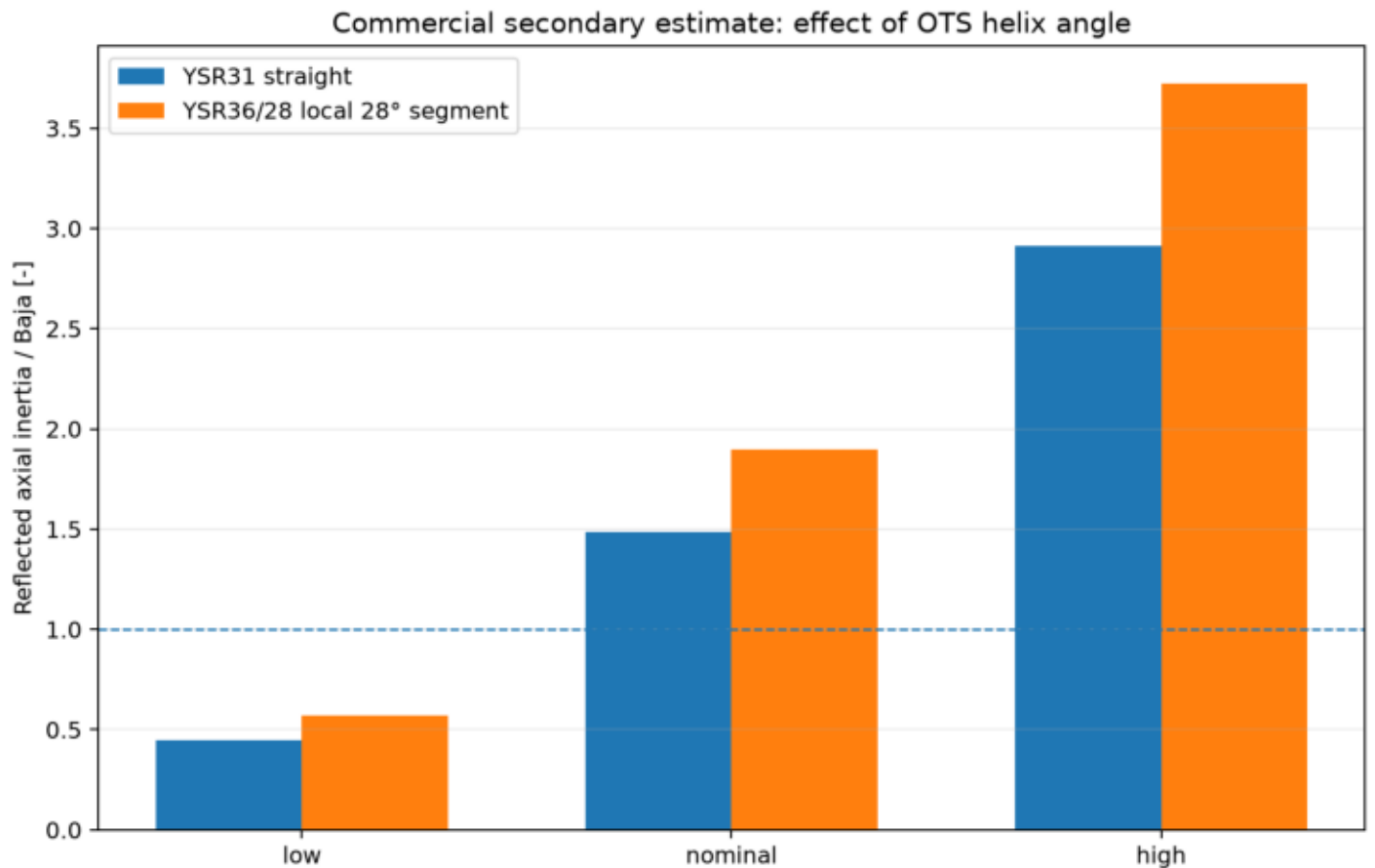
The screen selected the strongest clean continuous nominal case by preregistered rules. The same disturbance was then replayed on low / nominal / high commercial estimates.

System-level consequence under the selected disturbance

Estimate	Peak $P_{i,s,total}$	Peak $ \Delta F_{helix} [N]$	Max $ \Delta shift response $	Max $ \Delta primary rpm $	Max $ \Delta sec clamp [N]$
low	3.15%	31.2	0.109	15.4	40.2
nominal	7.87%	89.3	0.274	40.4	109.1
high	13.60%	162.4	0.454	68.5	189.6

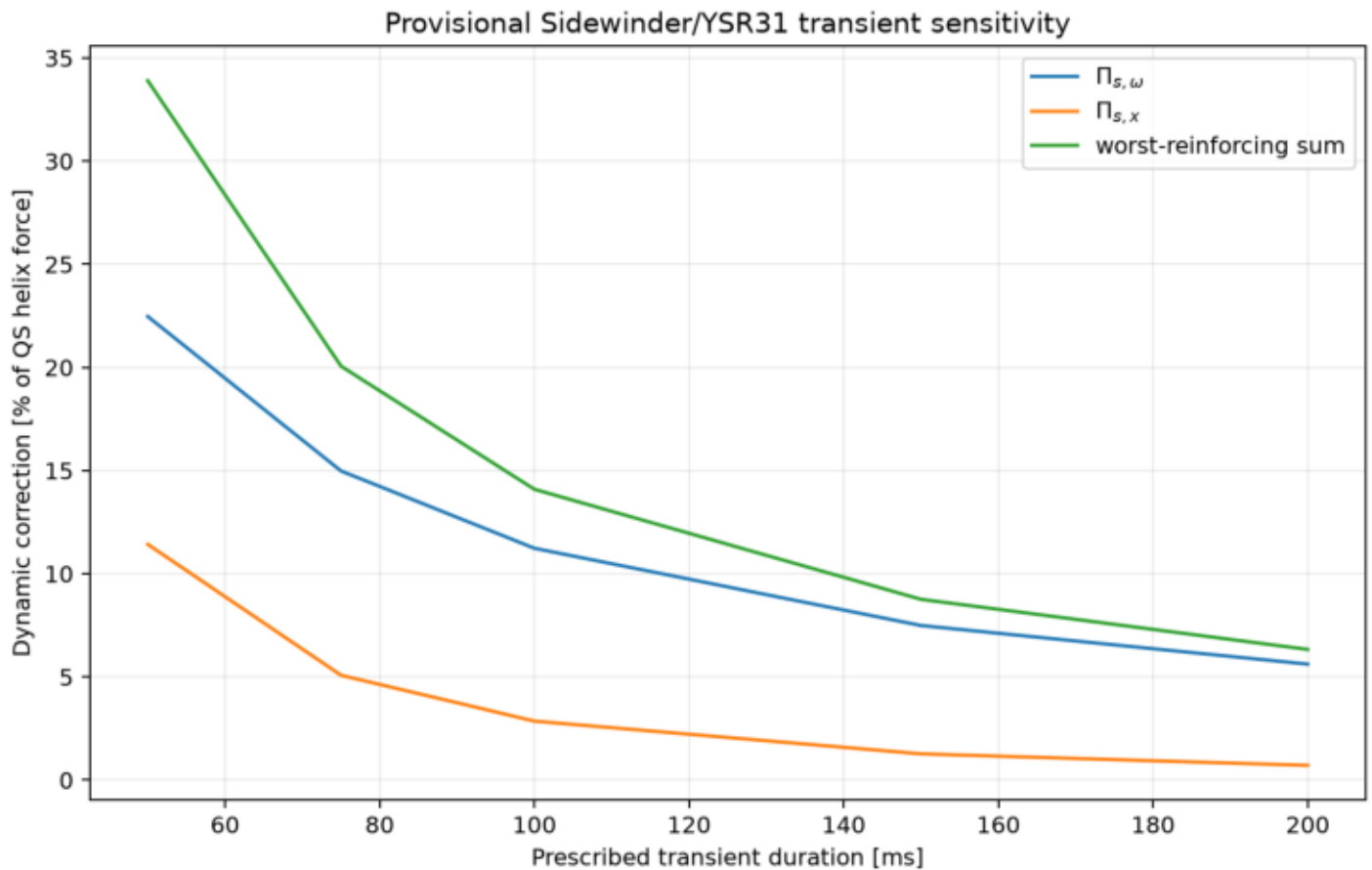
These are difference-in-differences metrics: the stressed response of the full-dynamic helix model is compared with the stressed response of the quasi-static helix model, after subtracting each model's own zero-disturbance control.

Commercial angle sensitivity vs Baja



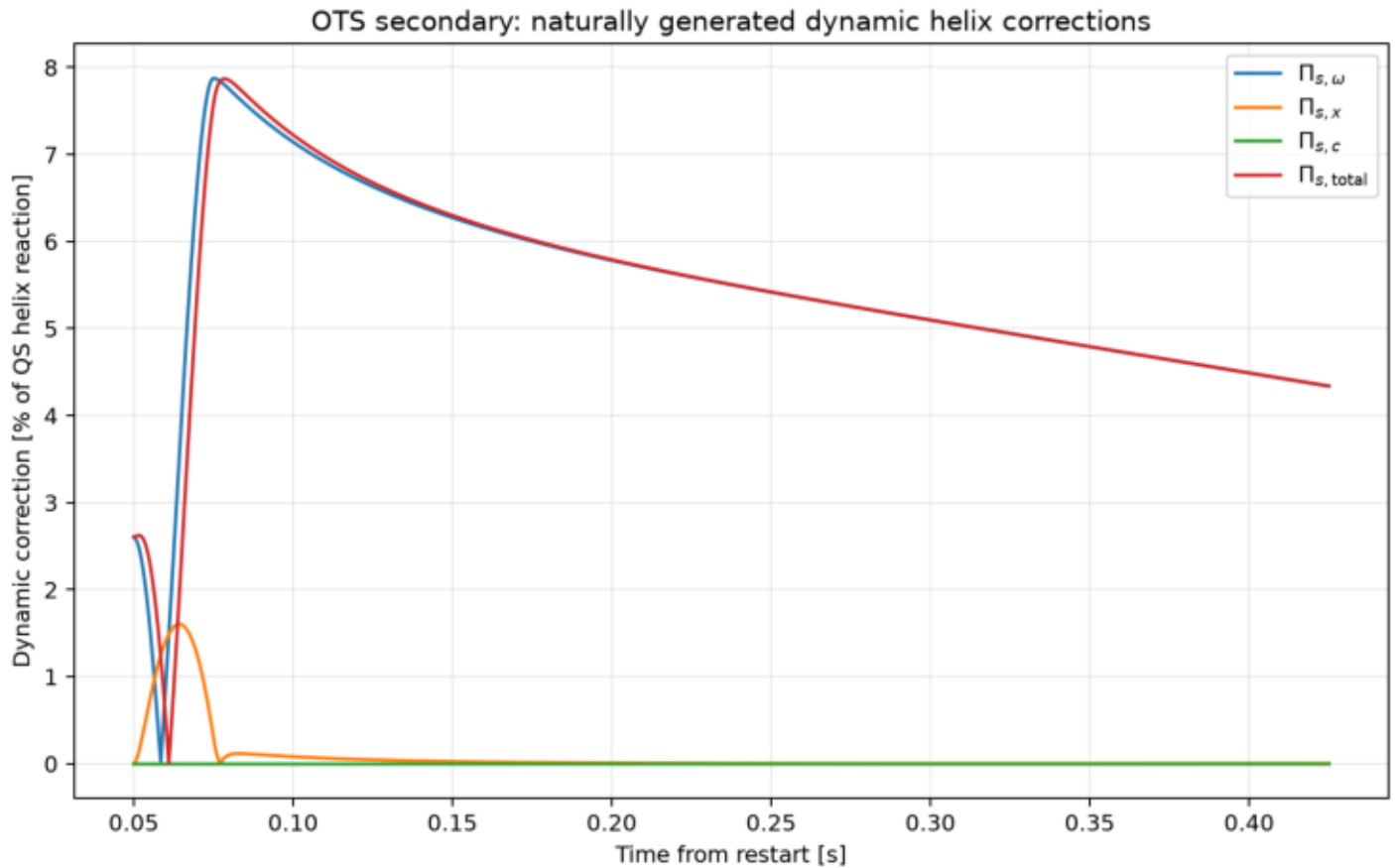
This figure shows why the commercial result is interesting. Even though the commercial helix is not as kinematically aggressive as the Baja baseline, plausible movable-member inertia can drive the reflected axial inertia well above Baja.

Prescribed transient sensitivity (component-level)



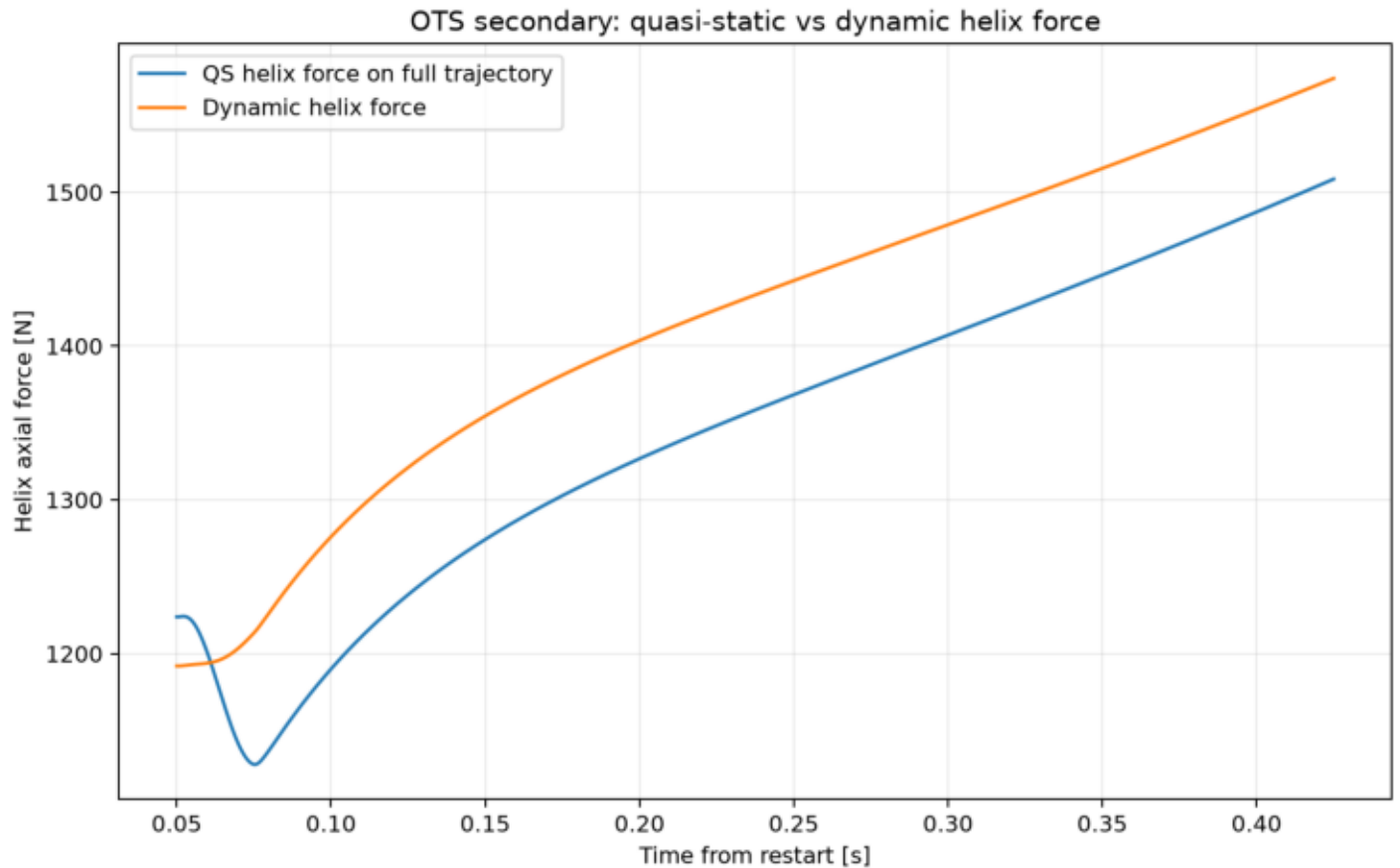
This was the component-level sensitivity precursor to the trajectory study. It shows that for the nominal hardware estimate, sufficiently fast transient scales push the exact dynamic correction into the tens-of-percent range. These are sensitivity inputs, not measured Sidewinder transients.

Nominal trajectory study – dynamic components



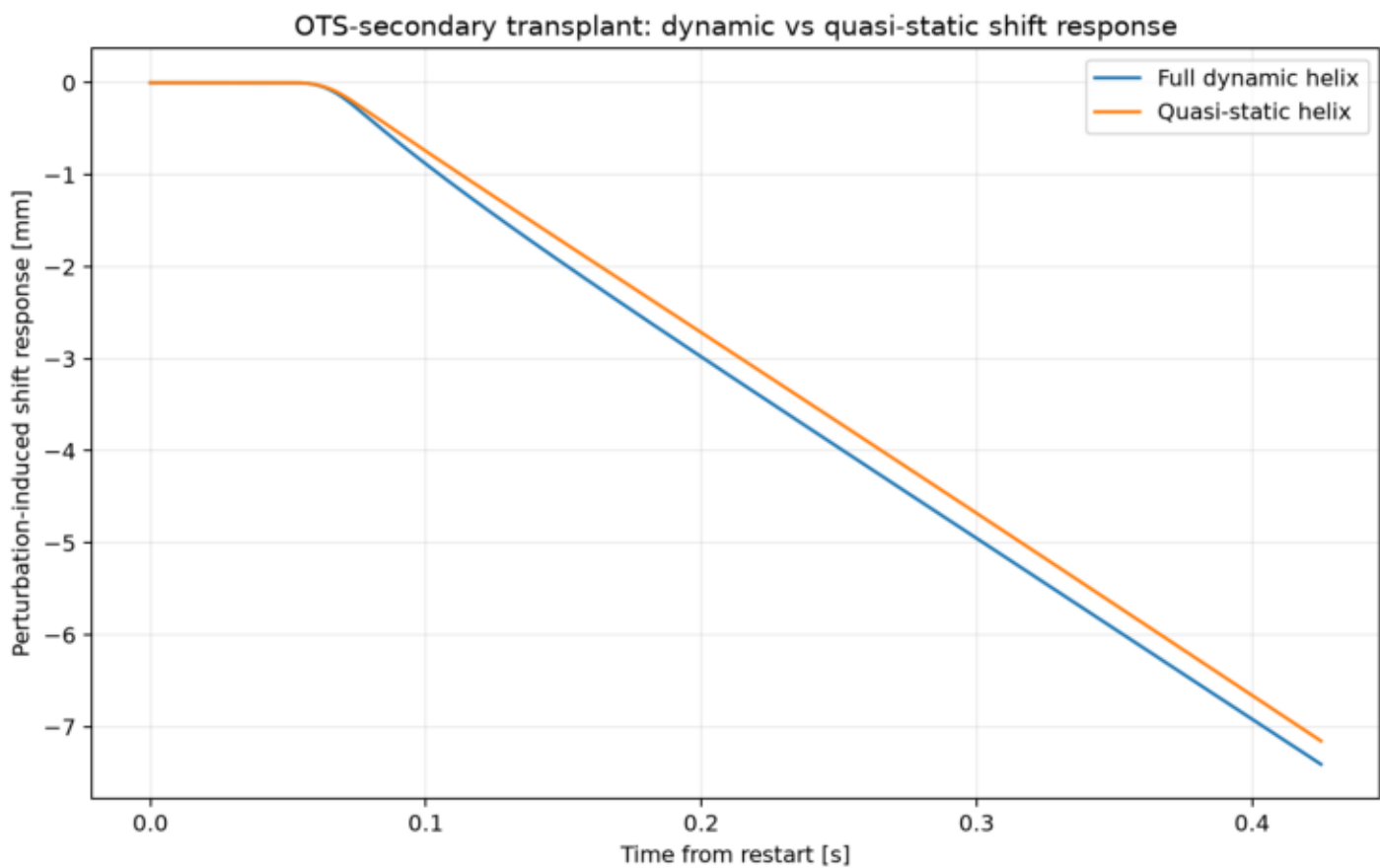
In the selected nominal trajectory case, the shaft-acceleration term dominates: peak $\Pi_{s,\omega} \approx 7.88\%$, while peak $\Pi_{s,x} \approx 1.60\%$ and the straight 31° helix has zero curvature term.

Nominal trajectory study – helix force consequence



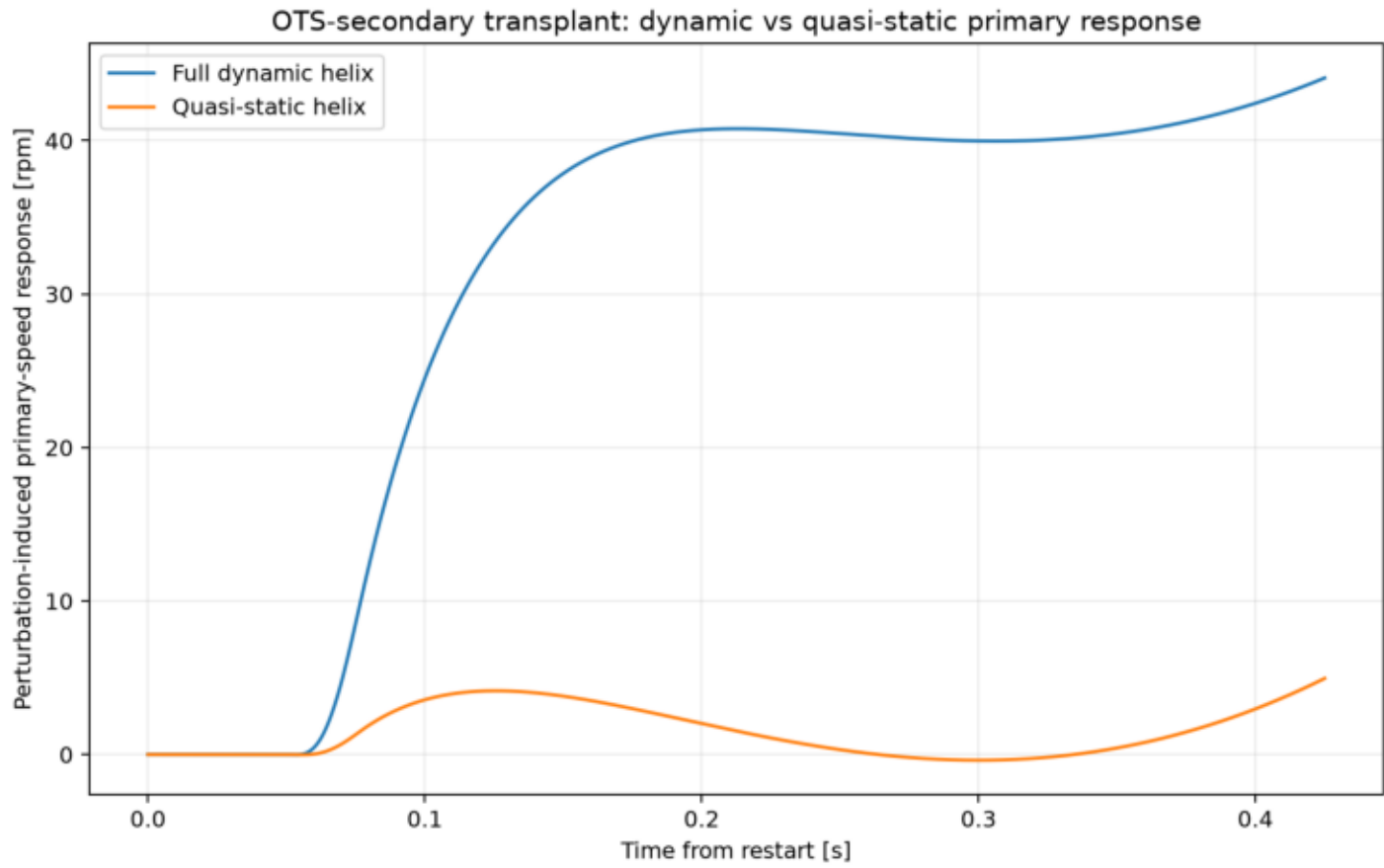
The selected nominal case produces a peak dynamic helix-force correction of about 89.3 N. This is a real internal load-path change, not a normalization artifact.

Nominal trajectory study — perturbation-induced shift response



The dynamic and quasi-static helix models generate visibly different shift responses under the same disturbance. The maximum difference is 0.274 mm, and at the time of maximum separation the response magnitude changes by roughly 6.3%.

Nominal trajectory study — perturbation-induced primary-speed response



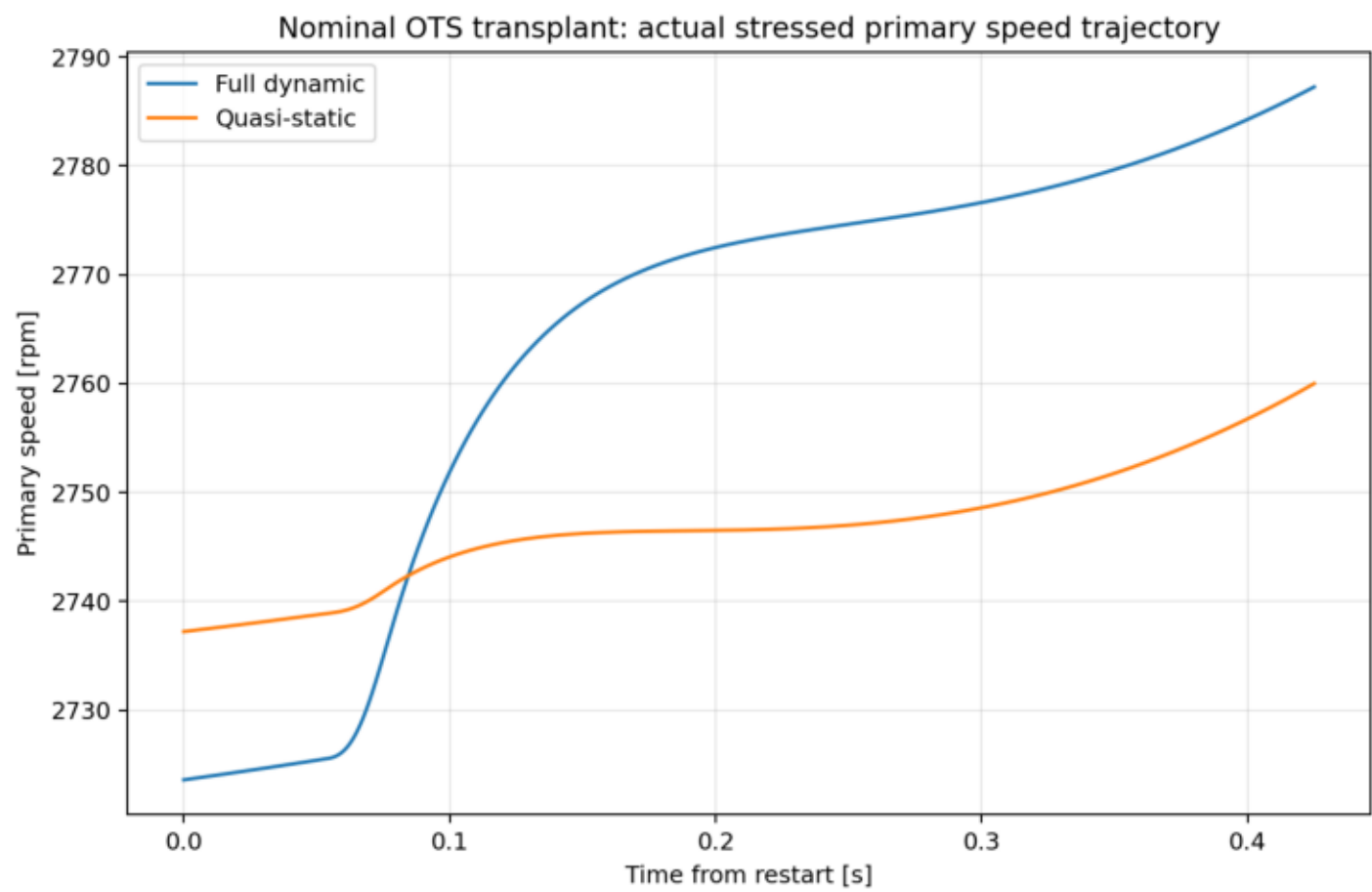
This is one of the clearest whole-system consequences. The selected nominal case produces a maximum dynamic-vs-QS primary-speed response difference of 40.4 rpm.

Nominal trajectory study – actual stressed shift trajectory



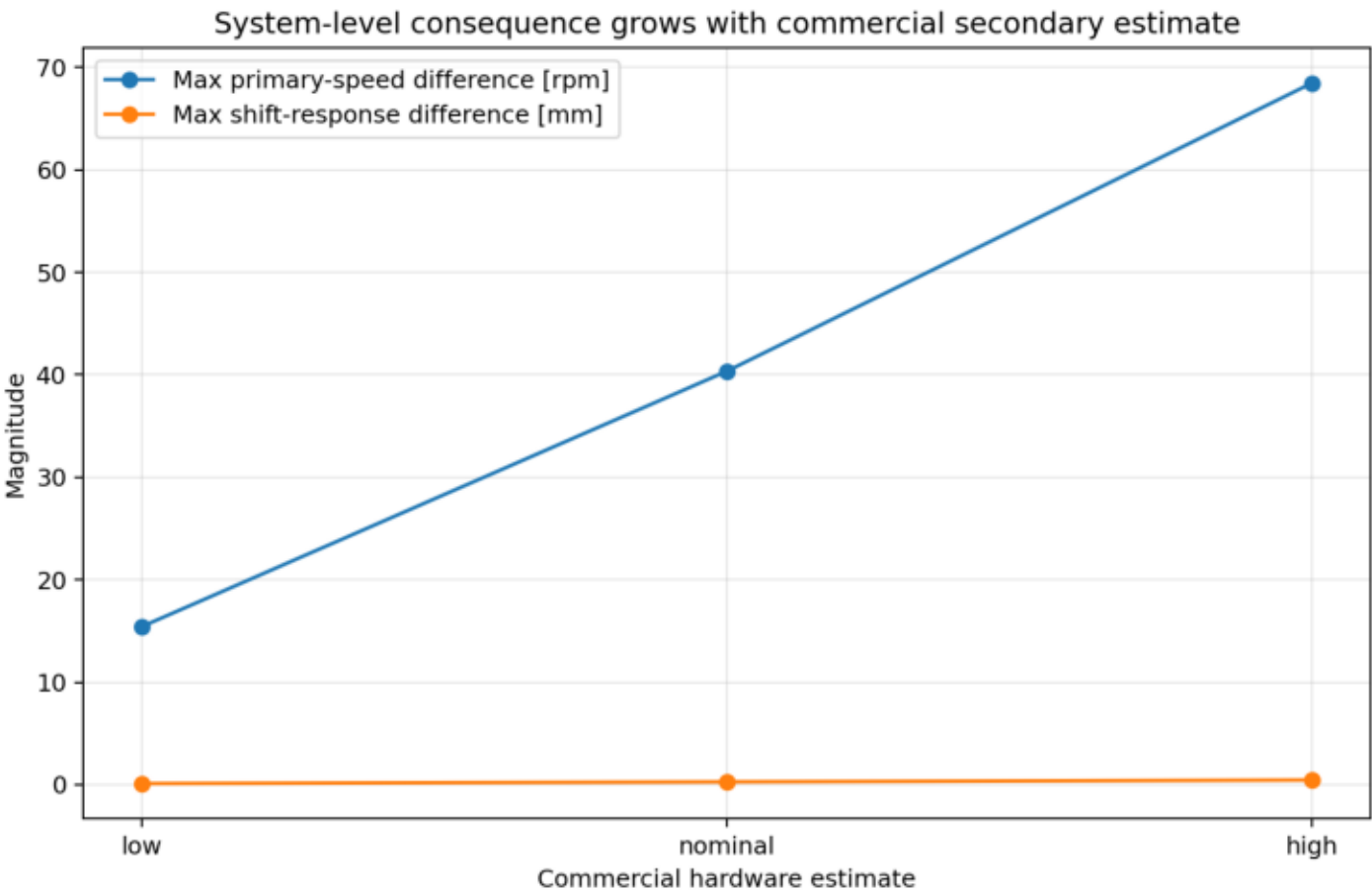
This plot shows the actual stressed trajectories rather than just the perturbation responses. The absolute trajectories remain close, which is expected, but the difference is systematic and not merely a plotting artifact.

Nominal trajectory study — actual stressed primary-speed trajectory



The stressed full-dynamic and quasi-static trajectories show a clear separation in primary speed during the transient. This complements the difference-in-differences response plot above.

Uncertainty trend across low / nominal / high estimates



As the provisional commercial mass-property estimate becomes more aggressive, both peak dynamic correction and whole-trajectory consequence grow monotonically. That is exactly the trend we hoped to see.

What to keep in mind when returning to this later

1. This actuator story is now scientifically coherent: (i) Baja ordinary operation is nearly quasi-static, (ii) the derivation identifies the exact dynamic terms and their scaling, (iii) OTS hardware can plausibly enlarge those terms, and (iv) a mechanism-transplant demonstration shows that the full CVT trajectory can change when the terms are no longer tiny.
2. The strongest current claim is existence / non-negligibility, not exact commercial-machine prediction. The paper language should stay disciplined on that boundary.
3. The most valuable future refinement is better commercial input quality, especially effective helix radius and movable-member rotational inertia. Those measurements control how far the nominal case truly sits between the current low and high bounds.
4. There is no urgent need to redesign these studies again right now. The output is strong enough to archive, cite in the results plan, and revisit once the other invariant and hybrid/transient studies are built out.